

Abdul Hadi

+92 300 0626647 | abdulhadi3049@gmail.com | LinkedIn | GitHub | Portfolio

Objective

BS Artificial Intelligence student at FAST with hands-on experience in computer vision, embedded systems, and software development. Seeking an AI/ML internship to apply practical skills in real-world systems while gaining industry exposure.

Education

National University of Computer and Emerging Sciences (FAST NUCES)
B.S. in Artificial Intelligence

Islamabad
Aug 2024 – Present

Experience

Lab Assistant · Part-time

FAST-NUCES, Islamabad | Jan 2026 – May 2026

Conducted weekly C++ OOP lab sessions for first-year Cybersecurity students, covering core concepts like classes, inheritance, and polymorphism. Graded assignments and resolved student queries.

Projects

NeoPet – PETiFI

Python, C++, TensorFlow Lite, MediaPipe, OpenCV, MQTT, ESP32

- Achieved approximately 95% face recognition accuracy using MobileFaceNet (TensorFlow Lite) and OpenCV, triggering intruder alerts (buzzer + red LEDs) with a QR code verification layer within a 3-second response latency.
- Enabled hands-free navigation by mapping 4 MediaPipe hand gestures to real-time motor commands, relayed from host laptop to ESP32 via MQTT for hardware execution.
- Resolved Python 3.12, MediaPipe, and Protobuf conflicts and voltage-induced ESP32 damage through iterative circuit protection, achieving stable real-time AI performance on constrained hardware.

EduVault

Node.js, Express.js, MySQL, JavaScript, HTML/CSS, JWT, bcryptjs, Multer

- Built a full-stack MVC platform with RESTful APIs enforcing role-based access control across 3 roles, domain-restricted auth, roll number validation via regex, and stateless JWT sessions with bcrypt-hashed passwords.
- Reduced content access friction by building a resource pipeline with Multer for multi-format uploads, an admin moderation workflow for resources and teacher accounts, audit-logged downloads, and in-browser PDF preview.
- Achieved API response times of 61-118ms across all core endpoints by implementing live keyword search, hierarchical subject/topic categorization, bookmarking, and a 5-star rating system on a MySQL backend.

AEGIS

Python, ROS2, YOLOv8, MediaPipe, ORB-SLAM, ESP32, UDP, Whisper

- Engineered an autonomous tactical rescue robot to navigate hazardous fire incident zones, track environmental threats, and evaluate situational risks under real-world constraints.
- Achieved a 95% classification accuracy in distinguishing conscious vs. unconscious victims under low-latency constraints by developing a synchronous vision pipeline combining a custom-trained YOLOv8 model with MediaPipe pose estimation.
- Streamlined hands-free commander updates with a 95% voice-command success rate by integrating a Whisper speech pipeline with a custom ORB-SLAM engine featuring native OpenGL hazard mapping and a low-latency ESP32 UDP data bridge.

Skills

- Programming Languages:** Python, C++, C, JavaScript, Arduino C
- AI, Computer Vision:** ROS2, YOLOv8, ORB-SLAM, MediaPipe, OpenCV, TensorFlow Lite (MobileFaceNet), Whisper, NumPy, Pandas, Matplotlib, Seaborn
- Web, Backend:** Node.js, Express.js, REST APIs, MySQL, JWT, bcryptjs, Multer, MySQL Workbench
- Developer Tools & Platforms:** Git, GitHub, OpenGL, VS Code, Arduino IDE, Jupyter Notebook

Certifications

- Robotics & Hardware Systems** – FAST-NUCES, Department of AI & DS
- Machine Learning Specialization** – Coursera, DeepLearning.AI

Volunteer Work

Student Volunteer · The Lighthouse Orphanage

Mar 2025 – May 2025

- Led an 8-week program with a 6-member team for underprivileged children, targeting UN SDGs 3, 4, and 15.
- Taught foundational topics in Computer Science (input/output), Math (DMAS), English grammar, and Science (digestive system/earthquakes).
- Managed language barriers and diverse learning speeds by adapting lessons on the spot for Punjabi, Pashto, and Balochi students.
- Executed co-curricular events, including an art competition, outdoor sports, and a plantation drive.